Supplementary Materials: Supplementary Materials for "SEMTRA: Global Semantic Transition and Rough-Set Rules for Auditable Post-hoc Explainability"

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1. Reproducibility Scope

This supplement documents the reviewer-driven v1 regeneration protocol. All claimable v1 values in the revised manuscript are copied from generated CSV, JSON, and LaTeX artifacts under `outputs/revision_v1/`. Raw datasets, downloaded archives, model weights, feature caches, and the local virtual environment are intentionally excluded from version control.

Table S1. Revision artifact index.

Location	Contents
`outputs/revision_v1/awa2/`	AwA2 dataset manifest, Protocol A seed-wise metrics, Protocol B seed-wise and per-class metrics, selected attributes, thresholds, rulebook, rule predictions, and WEDD-vs-MDLP paired statistics.
`outputs/revision_v1/sun/`	SUN xlsa17 manifest, transition metrics, selected attributes, thresholds, rulebook, and rule predictions.
`outputs/revision_v1/derm7pt/`	Derm7pt manifest, concept-transition metrics, rulebook, thresholds, and official-split test predictions.
`outputs/revision_v1/sensitivity/`	AwA2 selected-attribute count, SVD-rank, and confidence-threshold sensitivity grids.
`outputs/revision_v1/runtime/`	Per-stage runtime JSONL logs and summarized runtime CSV.
`outputs/revision_v1/statistics/`	Cross-domain claim-gating table used in the main manuscript.
`outputs/revision_v1/tables/`	LaTeX-ready tables generated from machine-readable artifacts.
`outputs/revision_v1/qc/`	Revision checklist artifacts used for consistency auditing.
`outputs/revision_v3/`	Enhanced prediction exports with true labels and base predictions, object-level bootstrap intervals, SUN hierarchy diagnostics, locked Derm7pt ResNet-50 features, v3 QC, and the shareable submission bundle.

2. Environment and Orchestration

Revision v1 is orchestrated by `scripts/run_revision_v1.py`. The runner accepts a repository root, dataset locations, seed lists, force-rebuild and smoke-test flags, and writes all outputs under `outputs/revision_v1/`. The full regeneration used seeds 42, 43, 44, 45, and 46 for AwA2. Revision v3 is orchestrated by `scripts/run_revision_v3.py` and writes enhanced prediction, SUN metadata, Derm7pt encoder, object-level interval, QC, and packaging artifacts under `outputs/revision_v3/`. The local environment was created as an ignored `.venv`; verified v3 imports included NumPy, pandas, SciPy, scikit-learn, matplotlib, pyarrow, h5py, Pillow, torch, and torchvision.

Table S2. Primary revision command and generated run summary.

Item	Value
Runner	`scripts/run_revision_v1.py`
Full command	`.venv\Scripts\python.exe scripts\run_revision_v1.py -seeds 42,43,44,45,46`
Run summary	`outputs/revision_v1/revision_v1_run_summary.json`
Environment manifest	`outputs/revision_v1/environment.json`
Status	`ok`; SUN and Derm7pt completed under the local data available for this revision.

3. Dataset Protocols

3.1. AwA2 and xlsa17

AwA2 uses the released 2048-dimensional ResNet-101 features distributed through the xlsa17 zero-shot benchmark resources [1]. Protocol A evaluates closed-world semantic reconstruction and rulebook audit metrics across the regenerated five-seed run. Protocol B follows the official xlsa17 seen/unseen split, training the transition on 40 seen classes and reporting unseen-class semantic-prototype transfer on 10 unseen classes. Protocol B is used as semantic-transfer validation only; it is not a competitive zero-shot learning leaderboard claim.

Table S3. AwA2 generated source files used for manuscript claims.

Artifact	Role
`awa2_protocol_a_seedwise.csv`	Per-seed transition, rulebook, coverage, fidelity, accuracy, abstention, and conflict metrics.
`awa2_protocol_a_summary.csv`	Mean and standard-deviation values reported in the abstract and Results.
`awa2_protocol_b_seedwise.csv`	Protocol B unseen-class prototype and symbolic-template transfer metrics.
`awa2_protocol_b_per_class_seed42.csv`	Per-class Protocol B diagnostic table.
`awa2_wedd_vs_mdlp_paired_stats.csv`	Paired WEDD-vs-MDLP statistics used to moderate the discretizer claim.

3.2. SUN Attribute Database

SUN uses the xlsa17 SUN ResNet-101 feature release and the 102-attribute SUN Attribute Database metadata [2,3]. The official SUN Attribute Database contains 14,340 images and 102 attributes over more than 700 scene categories; the xlsa17 release provides the feature and split representation used by the v1 runner. The reported SUN row states its scope explicitly because the low coverage and fidelity indicate that the current global symbolic template is weak for fine-grained scene categories.

Table S4. SUN xlsa17 semantic-transition and rulebook audit summary.

Scope	Objects	Classes	MAE	Coverage	Covered fidelity
Unseen prototype transfer	1440	72	0.0683	–	–
Seen-test rulebook audit	2580	645	–	0.0760	0.0918

Revision v3 normalizes the xlsa17 SUN image paths, joins them to the official SUN Attribute Database image list, and aggregates seen-test prediction behavior by scene hierarchy category. The resulting category diagnostics remain a stress-test artifact rather than a strengthened portability claim.

Table S5. SUN v3 low-coverage category diagnostics.

Category	Objects	Coverage	Conflict	Covered fidelity
abbey	4	0.0000	1.0000	–
access_road	4	0.0000	1.0000	–
airfield	4	0.0000	1.0000	–
airplane_cabin	4	0.0000	1.0000	–
airport/airport	4	0.0000	1.0000	–
airport/entrance	4	0.0000	1.0000	–

3.3. Derm7pt

Derm7pt uses dermoscopic images by default, official case-level train/validation/test index files, grouped diagnosis labels, and the seven checklist concepts [4–6]. Missing concept values are handled explicitly in the runner before model fitting. Revision v3 replaces the earlier handcrafted image-feature proxy with a locked TorchVision ResNet-50 ImageNet-1K v2 encoder applied to the `derm` image column in evaluation mode with no gradient updates. These results are retrospective technical-validation outputs only and must not be interpreted as clinical deployment evidence.

Table S6. Derm7pt retrospective technical-validation summary using official case-level splits.

Metric	Value
Test concept MAE	0.2433
Rulebook coverage	0.9241
Covered fidelity	0.5288
Covered accuracy	0.4219
Conflict rate	0.0051

Table S7. Derm7pt v3 retrospective technical-validation summary with locked ResNet-50 features.

Metric	Value
Encoder	TorchVision ResNet-50 ImageNet-1K v2
Feature dimension	2048
Test concept MAE	0.2359
Coverage	0.9899
Covered accuracy	0.5729
Covered fidelity	0.7417

4. Revision v3 Enhanced Prediction Exports

Revision v3 writes enhanced prediction CSV files for AwA2, SUN, and Derm7pt. Each row records the source object index, true label, true class, base-predictor decision, rulebook decision, abstention state, conflict state, activated rules, covered-correctness flag, and covered-fidelity flag. These fields make object-level confidence intervals reproducible without reconstructing hidden arrays from the runner.

Table S8. Revision v3 object-level bootstrap intervals from enhanced prediction exports.

Dataset	Metric	Mean	95% CI	Objects
AwA2	coverage	0.8469	[0.8394, 0.8547]	7465
AwA2	covered accuracy	0.4054	[0.3937, 0.4170]	7465
AwA2	covered fidelity to base	0.4109	[0.3990, 0.4228]	7465
AwA2	coverage	0.8257	[0.8171, 0.8342]	7465
AwA2	covered accuracy	0.3824	[0.3704, 0.3942]	7465
AwA2	covered fidelity to base	0.3949	[0.3828, 0.4064]	7465
AwA2	coverage	0.8926	[0.8856, 0.8991]	7465
AwA2	covered accuracy	0.4051	[0.3937, 0.4161]	7465
AwA2	covered fidelity to base	0.4070	[0.3948, 0.4189]	7465
AwA2	coverage	0.9027	[0.8966, 0.9094]	7465
AwA2	covered accuracy	0.4048	[0.3939, 0.4172]	7465
AwA2	covered fidelity to base	0.4128	[0.4010, 0.4244]	7465
AwA2	coverage	0.7721	[0.7625, 0.7823]	7465
AwA2	covered accuracy	0.3888	[0.3765, 0.4012]	7465
AwA2	covered fidelity to base	0.3985	[0.3855, 0.4113]	7465
SUN	coverage	0.0760	[0.0659, 0.0868]	2580
SUN	covered accuracy	0.0714	[0.0389, 0.1083]	2580
SUN	covered fidelity to base	0.0918	[0.0511, 0.1313]	2580
Derm7pt	coverage	0.9899	[0.9797, 0.9975]	395
Derm7pt	covered accuracy	0.5729	[0.5229, 0.6218]	395
Derm7pt	covered fidelity to base	0.7417	[0.6964, 0.7863]	395

5. Sensitivity and Runtime Diagnostics

The v1 runner regenerates sensitivity grids for the number of selected attributes q, SVD rank r, and minimum rule-confidence thresholds. The manuscript includes the q and r summaries because these parameters directly affect rulebook compactness and coverage. The threshold frontier is retained as a machine-readable artifact to support follow-up auditing.

Table S9. Selected-attribute-count sensitivity on AwA2 for the WEDD rulebook.

q	Coverage	Covered fidelity	Covered accuracy	Conflict	Rules
8	0.7999	0.3435	0.3422	0.1711	59
12	0.8251	0.4002	0.3889	0.1522	60
18	0.8469	0.4109	0.4054	0.1378	57
24	0.8354	0.4343	0.4266	0.1365	54
32	0.8719	0.5522	0.5380	0.1027	50

Table S10. SVD-rank sensitivity on AwA2 for semantic reconstruction and rulebook audit metrics.

Rank	MAE	Coverage	Covered fidelity	Covered accuracy	Runtime s
16	0.1295	0.9037	0.3875	0.3913	7.4
32	0.1150	0.9023	0.3668	0.3695	7.2
64	0.1031	0.8469	0.4109	0.4054	6.8
128	0.0942	0.7893	0.4898	0.4801	6.8

Table S11. Revision-v1 per-phase runtime summary from JSONL instrumentation.

Dataset	Phase	Runs	Mean s	Total s
AwA2	protocol_a_Equal frequency	5	5.5	27.3
AwA2	protocol_a_Equal width	5	5.3	26.6
AwA2	protocol_a_MDLP-like entropy	5	7.3	36.6
AwA2	protocol_a_WEDD	5	7.2	36.0
AwA2	protocol_b_semantic_validation	5	3.0	15.1
AwA2	q_sensitivity	5	7.0	34.9
AwA2	svd_rank_sensitivity	4	7.1	28.3
Derm7pt	q_sensitivity	3	0.1	0.3
Derm7pt	transition_and_rulebook	1	7.8	7.8
SUN	q_sensitivity	6	22.0	132.0
SUN	transition_and_rulebook	1	11.6	11.6

6. Rule Traces and Perturbation Diagnostics

The v1 artifact set preserves the seed-42 AwA2 rulebook, rule predictions, WEDD thresholds, and selected attributes so that individual rule traces can be reconstructed from source data. The main manuscript includes representative traces and perturbation diagnostics; the generated CSV files are the authoritative audit trail for reproducing matched antecedents, rule supports, confidences, and abstention cases. Future revisions should regenerate these traces after any change to the selected concept set, discretizer, rule confidence threshold, or soft-matching radius.

7. Claim Gating

Cross-domain claims are gated by completed, traceable artifacts. AwA2, SUN, and Derm7pt are included because their local runs completed and wrote CSV/JSON/LaTeX artifacts. Revision v3 improves traceability by exporting true labels and base predictions with every prediction row, but it does not change the scope of the claims. The Derm7pt wording is intentionally restricted to retrospective technical validation. SUN and Derm7pt subset or protocol limitations must be stated wherever their results are summarized.

References

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